

Chapter 7, Problem 89CP

Problem

An RL circuit may be used as a differentiator if the output is taken across the inductor and $\tau \ll T$ (say $\tau < 0.1T$), where T is the width of the input pulse. If R is fixed at $200\text{ k}\Omega$, determine the maximum value of L required to differentiate a pulse with $T = 10\text{ }\mu\text{s}$.

Step-by-step solution

Step 1 of 1

Since $\tau < 0.1T = 1\text{ }\mu\text{s}$.

$$\frac{L}{R} < 1\text{ }\mu\text{s}$$

$$L < R \times 10^{-6} = (200 \times 10^3)(1 \times 10^{-6})$$

$L < 200\text{ mH}$.